

City Prioritization For Air Filtration Market Entry

EPA Air Quality · NOAA Climate · U.S. Census Population

Los Angeles · New York · Houston · Phoenix · Chicago



BUSINESS PROBLEM

WHAT PROBLEM ARE WE SOLVING?

An industrial air filtration company needs to decide which U.S. cities to enter first. Rather than relying on intuition, leadership wants a data-driven city prioritization strategy → identifying where demand for filtration products will be strongest over the next 12 months.

The Core Question

Which of five major U.S. cities should be prioritized for initial product launch and sales investment?

Who Is This For?

Sales and market entry leadership who need a clear, evidence-based ranking of target cities.

Our Approach

Combine BI/EDA with predictive modeling to produce a Market Opportunity Score for each city.

Three Datasets, One Decision

EPA Air Quality System (AQS)

Pollution

Daily records for PM2.5, O3, CO, and NO2 across five cities, 2015–2024. ~66,000 daily records. Primary signal for pollution burden and health risk.

→ Establishes which cities have the most persistent and severe air quality problems.

U.S. Census Bureau

Population

Annual city-level population estimates. Captures year-over-year growth and reflects increasing congestion, vehicle use, and energy demand.

→ Population scales exposure → a high-pollution city with more residents means more potential customers.

NOAA Climate Data (CDO)

Climate

Monthly average temperature and precipitation from local airport weather stations — the most consistent source available. 2015–2024.

→ Heat accelerates ozone and particle formation; rain scrubs particles. Climate tells us which cities get natural relief and which don't.

Why Climate Data Strengthens the Business Case

Temperature Effects

PM2.5

Heat increases wildfire risk and drives secondary particle formation through atmospheric chemistry.

O3 / Ozone

Most directly affected → hot sunny days produce significantly more ground-level ozone.

NO2

Higher temps accelerate chemical reactions that form NO2 from NOx emissions.

CO

Slightly increases combustion efficiency; hot weather also drives more vehicle and industrial activity.

Precipitation Effects

PM2.5

Most directly affected → rain physically scrubs fine particles from the air, causing sharp drops after rainfall.

O3 / Ozone

Rainy days are cloudier, reducing sunlight and slowing ozone formation.

NO2

Rain washes NO2 out of the atmosphere, temporarily reducing concentrations.

CO

Minimal direct effect → CO does not dissolve easily in water.

Key Insight: Cities with hot, dry climates like Los Angeles and Phoenix are structurally more prone to elevated PM2.5 and ozone → natural climate offers no pollution relief, strengthening the market entry case for those cities.

COMBINED DATASET

DATA SOURCES - FEATURE DATASET										
Final Merged Dataset — Sample (600 rows x 10 features)										
City - Monthly granularity - EPA AQS + NOAA Climate + U.S. Census - 2015-2024										
										[Target variable]
	city	year	month	precip_in	tavg_f	yr_pop	CO	NO2	O3	PM2.5
0	Chicago	2015	1	1.41	22.4	2757203	0.239951	19.555521	0.016170	13.280614
1	Chicago	2015	2	1.46	14.7	2757203	0.347014	20.637091	0.021669	13.200935
2	Chicago	2015	3	1.10	35.4	2757203	0.259718	19.284992	0.026579	12.673560
3	Chicago	2015	4	2.87	49.4	2757203	0.271056	15.979784	0.030318	9.220072
4	Chicago	2015	5	4.67	60.3	2757203	0.289392	13.292322	0.034010	9.321706
5	Chicago	2015	6	7.12	67.3	2757203	0.243698	13.842576	0.032270	9.548336
6	Chicago	2015	7	2.85	72.3	2757203	0.375038	12.422307	0.033153	14.741716
7	Chicago	2015	8	2.16	71.7	2757203	0.529688	10.659230	0.029487	10.031705
8	Chicago	2015	9	4.64	69.0	2757203	0.192319	12.569572	0.029884	9.920425
9	Chicago	2015	10	2.24	54.7	2757203	0.150814	12.270344	0.021399	7.697172
10	Chicago	2015	11	4.49	44.6	2757203	0.238740	16.248739	0.019450	9.297613
11	Chicago	2015	12	4.87	39.0	2757203	0.228762	13.698197	0.016210	9.425732
12	Chicago	2016	1	0.85	24.8	2755833	0.244532	18.564831	0.018238	9.512129
13	Chicago	2016	2	1.23	30.5	2755833	0.234720	17.652560	0.022573	8.601076
14	Chicago	2016	3	3.35	43.3	2755833	0.240124	15.320639	0.024899	7.682656
15	Chicago	2016	4	2.80	47.8	2755833	0.209121	15.473370	0.033819	8.862459
16	Chicago	2016	5	5.43	59.7	2755833	0.204032	15.937565	0.034247	7.988585
17	Chicago	2016	6	2.85	71.6	2755833	0.189118	14.794212	0.039091	8.665418
18	Chicago	2016	7	6.24	75.5	2755833	0.189885	11.868838	0.034181	9.438965
19	Chicago	2016	8	4.26	75.8	2755833	0.213336	11.512378	0.030564	8.584532

Four Pollutants — One Primary Focus

PM2.5

★ PRIMARY FOCUS

Fine Particulate Matter

Fine particles ≤ 2.5 microns from vehicle exhaust, industrial emissions, wildfires, and construction dust.

★ Primary Health Indicator

O3

Ground-Level Ozone

Forms when NO_x and VOCs react in sunlight. Most directly driven by hot, sunny days → summer spikes are highly consistent.

CO

Carbon Monoxide

Combustion byproduct from vehicles, power plants, and industrial machinery. Indicator of traffic and industrial activity.

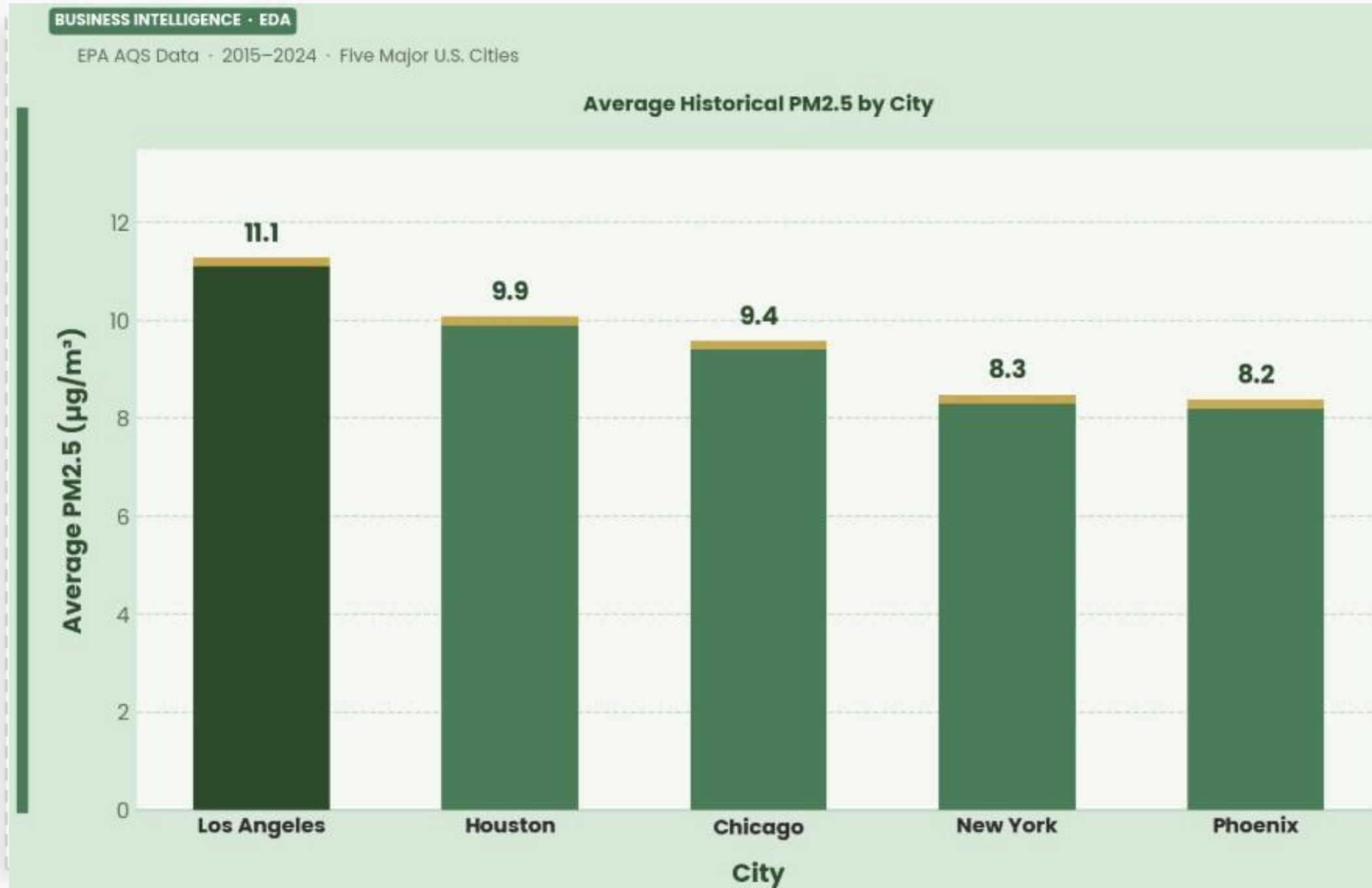
NO2

Nitrogen Dioxide

Traffic and industrial emissions marker. Precursor to smog and acid rain. Higher temps accelerate its formation.

PM2.5 is the most detrimental to human health and serves as our primary modeling target.

Historical PM2.5 by City: Who Carries the Greatest Burden?



Los Angeles

Consistently highest average PM2.5 → driven by traffic density, industrial activity, and hot dry climate.

Houston

Elevated levels tied to petrochemical industry and high vehicle use.

Phoenix

Strong PM2.5 burden amplified by desert heat and low precipitation.

New York

Moderate pollution but massive population exposure scales the total burden.

Chicago

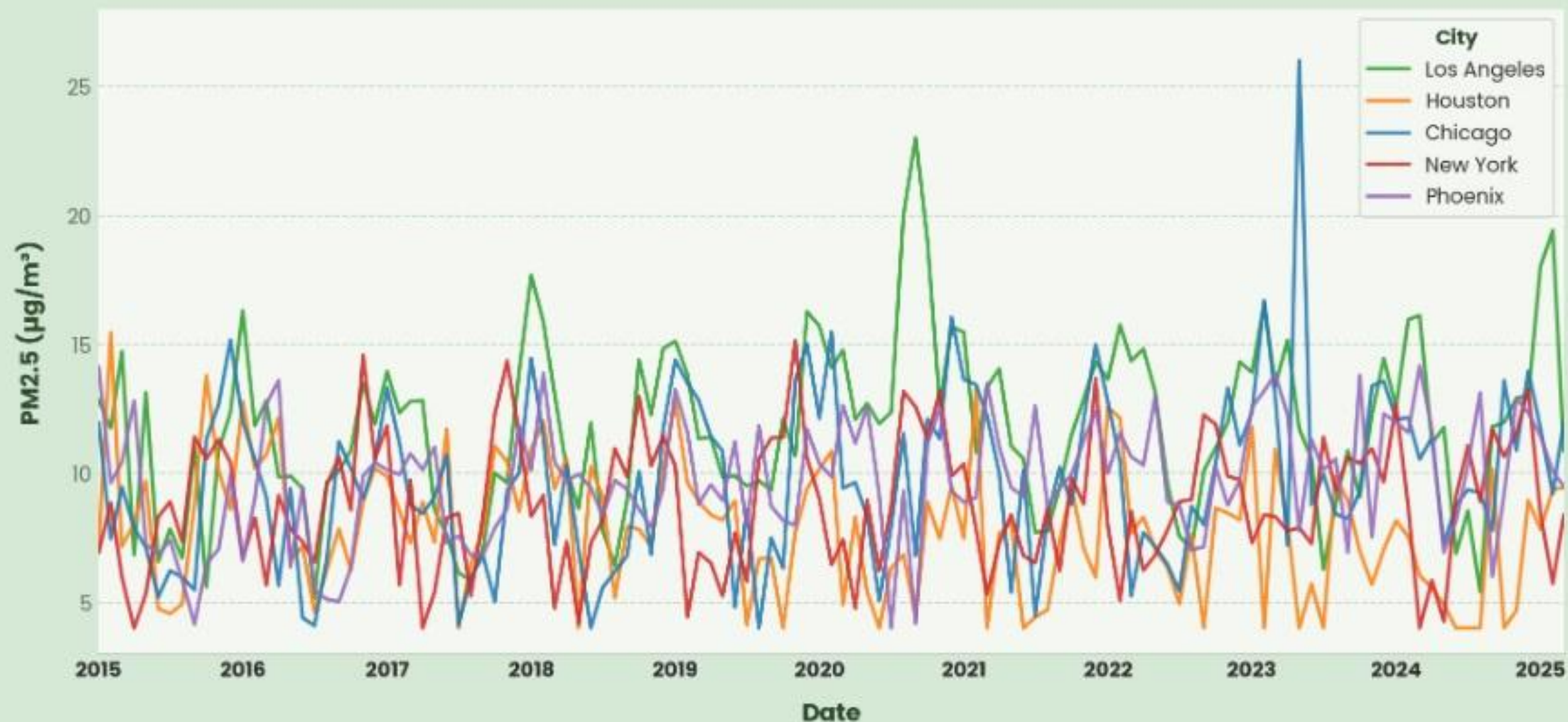
Lowest average PM2.5 among the five cities in historical data.

PM2.5 Trends Over Time: Patterns That Shape Our Forecast

BUSINESS INTELLIGENCE · EDA

EPA AQS Data · 2015–2025 · Monthly Average PM2.5 ($\mu\text{g}/\text{m}^3$)

Historical PM2.5 Trends by City



Strong Seasonality

PM2.5 peaks in winter months across all cities → a pattern our model explicitly encodes using sine/cosine features.

Historical Persistence

Cities with high PM2.5 in prior years tend to stay high → past 12-month averages are among the strongest predictors of future burden.

DATA SCIENCE & PREDICTIVE MODELING



PREDICTIVE MODELING APPROACH



Our predictive pipeline combined pollution, climate, and population data into a monthly forecasting dataset covering five U.S. cities from 2015 to 2024.

To preserve real-world forecasting conditions, we used a time-based split rather than random splitting. The model was trained on earlier time periods and evaluated on the most recent 12 months of data.

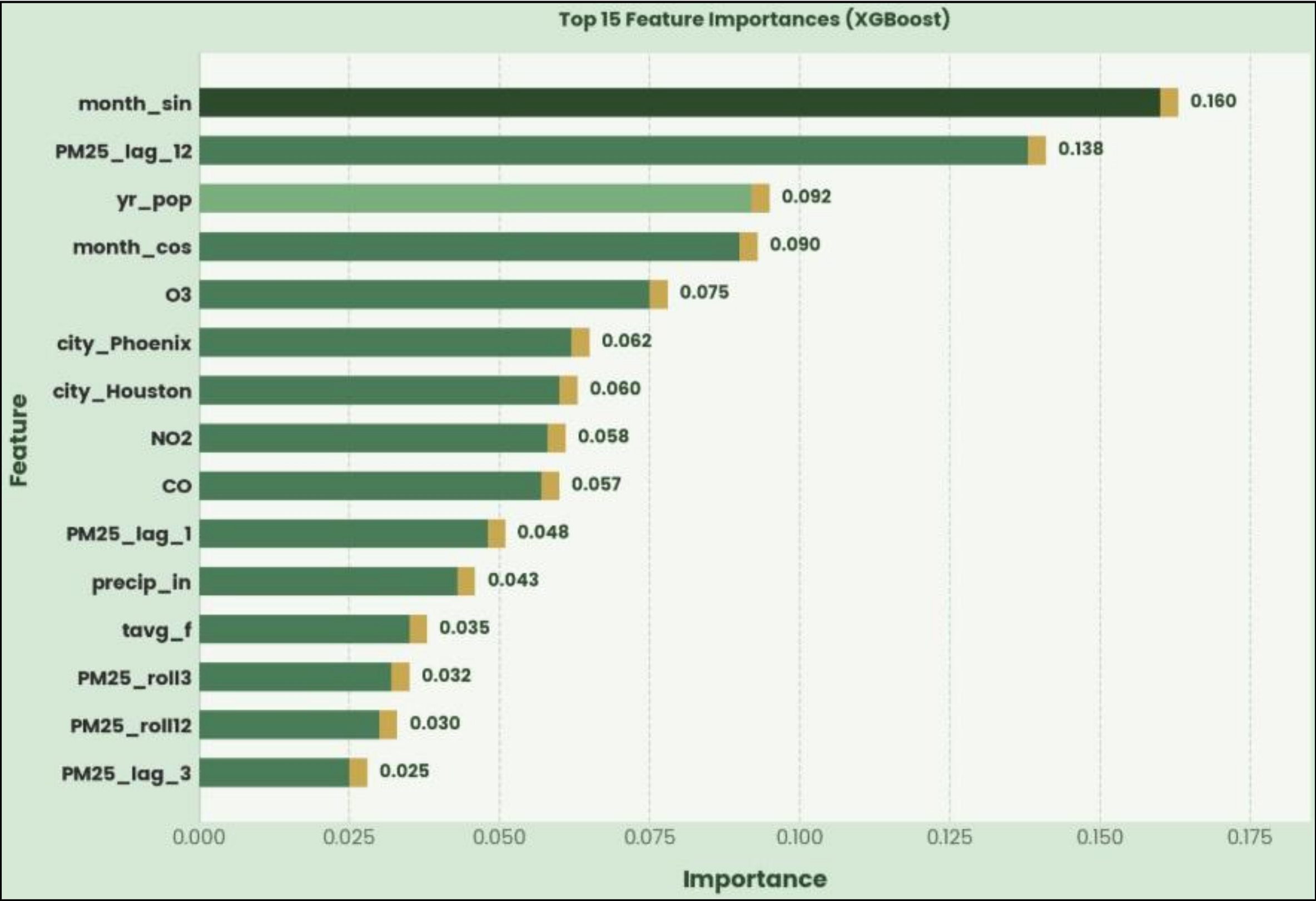
We compared multiple machine learning models, including Linear Regression, Random Forest, and XGBoost, to predict future PM2.5 levels. Based on model performance, XGBoost was selected as the final forecasting model because it captured non-linear pollution patterns and seasonal variation more effectively than the baseline models.

The final model was then used to generate a 12-month PM2.5 forecast outlook for each city. Performance was evaluated using MAE, RMSE, and R-squared.

Model	RMSE	R ²
Linear Regression	2.34	0.45
Random Forest	1.91	0.64
XGBoost	1.83	0.67



FEATURE IMPORTANCE



FORECAST RESULTS



THE 12-MONTH FORECAST RESULTS

MAIN INSIGHTS

INSIGHT #1

XGBoost delivered the strongest predictive performance.

It was selected as the final model because it achieved the lowest RMSE and highest R^2 among the tested models.

INSIGHT #2

Los Angeles remains the highest-risk city in the forecast horizon

Its predicted PM2.5 burden stays higher than other cities, making it a strong candidate for priority market entry.

INSIGHT #3

Forecast differences support city prioritization. Cities differ in projected burden, persistence, and volatility, so launch sequencing should be based on both environmental risk and market exposure.

BUSINESS OUTPUT & & MARKET OPPORTUNITY SCORE



From Forecast to Business Decision

$$\text{Market Opportunity Score} = 0.40 \times \text{Pollution} + 0.25 \times \text{Population} + 0.20 \times \text{High-Risk Months} + 0.15 \times \text{Persistence}$$

40%

Predicted Avg PM2.5

12-month forecasted pollution level → primary demand signal.

25%

Population Exposure

Projected population → scales market size.

20%

High-Risk Months

Months forecast above 12 $\mu\text{g}/\text{m}^3$ threshold → urgency indicator.

15%

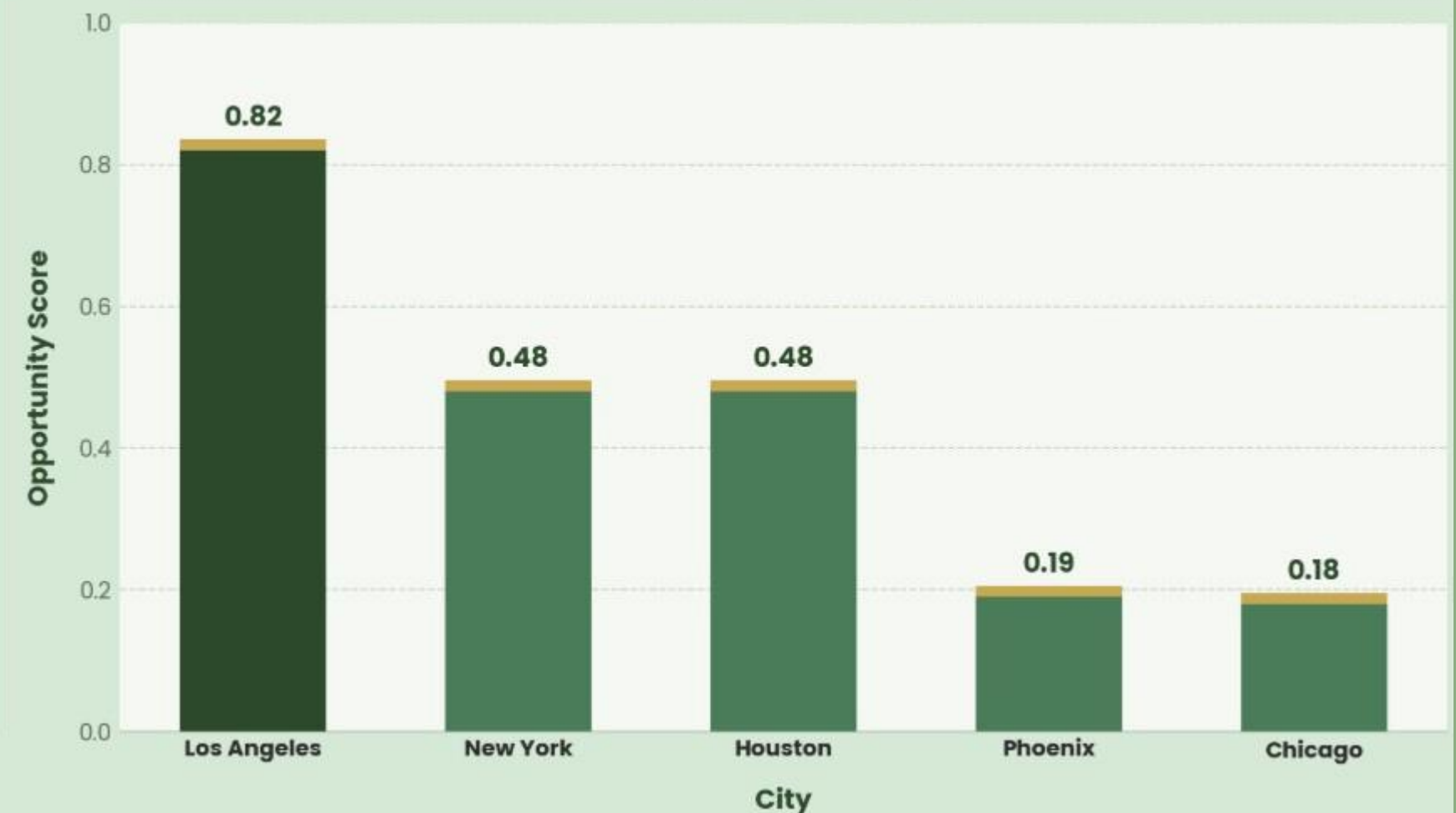
Historical Persistence

Last 12 months of actual PM2.5 → confirms structural problem.

BUSINESS OUTPUT · MARKET OPPORTUNITY

Weighted composite: PM2.5 Forecast · Population · High-Risk Months · Persistence

Market Opportunity Score by City



City Launch Priority: Final Rankings

1

Los Angeles

Highest forecasted PM2.5, multiple high-risk months, strong historical persistence.

Top priority → pollution burden + persistent structural risk.

2

New York

Moderate forecast PM2.5 but largest population exposure of the five cities.

Scale of exposure makes it a strong #2 despite lower pollution intensity.

3

Houston

Elevated PM2.5 tied to petrochemical industry. Moderate population.

Strong pollution story; Phase 2 target.

4

Phoenix

High-pollution months driven by desert heat. Smaller population than LA/NY.

Climate-driven risk is real; smaller market scales the score down.

5

Chicago

Lowest forecasted PM2.5 among the five. Declining historical trend.

Lowest opportunity score → deprioritize for initial launch.

Prioritize. Launch. Expand.

01

Launch First in Los Angeles

Highest forecasted PM2.5, structural climate risk, and strong historical persistence. Clearest ROI for first-mover investment.

02

New York as Phase 1B

Population exposure scales market size significantly. Even moderate pollution intensity translates to broad demand.

03

Houston & Phoenix: Phase 2

Real pollution burden but smaller addressable market. Expand after establishing West Coast and Northeast presence.

04

Reassess Annually

Rankings should be refreshed each year as population, climate conditions, and emission trends evolve.

We built a predictive model to estimate future city-level pollution burden, then translated those forecasts into a market-prioritization tool for leadership decision-making.



THANK YOU!